Complex Analysis Homework 6

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6. Give another proof of Lemma 1.3 by showing directly that the Laplacian of $u \circ F$ is zero. [Hint: The real and imaginary parts of F satisfy the Cauchy-Riemann equations.]

证明 直接计算。

设

$$F = \xi + i\eta, \quad u \circ F = u\big(\xi(x,y), \eta(x,y)\big),$$

其中 $\dfrac{\partial^2 u}{\partial \xi^2} + \dfrac{\partial^2 u}{\partial \eta^2} = 0$ 以及 $\dfrac{\partial \xi}{\partial x} = \dfrac{\partial \eta}{\partial y}$, $\dfrac{\partial \eta}{\partial x} = -\dfrac{\partial \xi}{\partial y}$, 再次求偏导有

$$\frac{\partial^2 \xi}{\partial x^2} = \frac{\partial^2 \eta}{\partial x \partial y} = -\frac{\partial^2 \xi}{\partial y^2}, \quad \frac{\partial^2 \eta}{\partial x^2} = -\frac{\partial^2 \xi}{\partial x \partial y} = -\frac{\partial^2 \eta}{\partial y^2}.$$

从而

$$\frac{\partial^2}{\partial x^2}(u \circ F) = \frac{\partial^2 u}{\partial \xi^2}\left(\frac{\partial \xi}{\partial x}\right)^2 + \frac{\partial u}{\partial \xi}\frac{\partial^2 \xi}{\partial x^2} + 2\frac{\partial^2 u}{\partial \xi \partial \eta}\frac{\partial \xi}{\partial x}\frac{\partial \eta}{\partial x} + \frac{\partial^2 u}{\partial \eta^2}\left(\frac{\partial \eta}{\partial x}\right)^2 + \frac{\partial u}{\partial \eta}\frac{\partial^2 \eta}{\partial x^2}$$

$$\frac{\partial^2}{\partial y^2}(u \circ F) = \frac{\partial^2 u}{\partial \xi^2}\left(\frac{\partial \xi}{\partial y}\right)^2 + \frac{\partial u}{\partial \xi}\frac{\partial^2 \xi}{\partial y^2} + 2\frac{\partial^2 u}{\partial \xi \partial \eta}\frac{\partial \xi}{\partial y}\frac{\partial \eta}{\partial y} + \frac{\partial^2 u}{\partial \eta^2}\left(\frac{\partial \eta}{\partial y}\right)^2 + \frac{\partial u}{\partial \eta}\frac{\partial^2 \eta}{\partial y^2},$$

相加即得。

7. Provide all the details in the proof of the formula for the solution of the Dirichlet problem in a strip discussed in Section 1.3. Recall that it suffices to compute the solution at the points $z = iy$ with $0 < y < 1$.

(a) Show that if $re^{i\theta} = G(iy)$, then

$$re^{i\theta} = i\frac{\cos \pi y}{1 + \sin \pi y}.$$

This leads to two separate cases: either $0 < y \leq 1/2$ and $\theta = \pi/2$, or $1/2 \leq y < 1$ and $\theta = -\pi/2$. In either case, show that

$$r^2 = \frac{1 - \sin \pi y}{1 + \sin \pi y} \quad \text{and} \quad P_r(\theta - \varphi) = \frac{\sin \pi y}{1 - \cos \pi y \sin \varphi}.$$

(b) In the integral $\dfrac{1}{2\pi}\displaystyle\int_0^\pi P_r(\theta - \varphi)\tilde{f}_0(\varphi)d\varphi$ make the change of variables $t = F(e^{i\varphi})$. Observe that

$$e^{i\varphi} = \frac{i - e^{\pi t}}{i + e^{\pi t}},$$

and then take the imaginary part and differentiate both sides to establish the two identities

$$sin\varphi = \frac{1}{\cosh \pi t} \quad \text{and} \quad \frac{d\varphi}{dt} = \frac{\pi}{\cosh \pi t}.$$

Hence deduce that

$$\begin{aligned}\frac{1}{2\pi}\int_0^{\pi} P_r(\theta-\varphi)\tilde{f}_0(\varphi)d\varphi &= \frac{1}{2\pi}\int_0^{\pi}\frac{\sin \pi y}{1-\cos \pi y \sin \varphi}\tilde{f}_0(\varphi)d\varphi \\ &= \frac{\sin \pi y}{2}\int_{-\infty}^{\infty}\frac{f_0(t)}{\cosh \pi t - \cos \pi y}dt.\end{aligned}$$

(c) Use a similar argument to prove the formula for the integral $\frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta-\varphi)\tilde{f}_1(\varphi)d\varphi$.

证明 *(a)* 代入 $z = iy$,

$$G(iy) = \frac{i - e^{i\pi y}}{i + e^{i\pi y}} = \frac{-\cos \pi y + i(1 - \sin \pi y)}{\cos \pi y + i(1 + \sin \pi y)}.$$

分子分母同乘共轭并化简，得

$$G(iy) = i\,\frac{\cos \pi y}{1 + \sin \pi y}.$$

记 $G(iy) = re^{i\theta}$，则 $\theta = \pm\pi/2$，且

$$r^2 = \frac{\cos^2 \pi y}{(1 + \sin \pi y)^2} = \frac{1 - \sin \pi y}{1 + \sin \pi y}.$$

泊松核

$$P_r(\theta - \varphi) = \frac{1 - r^2}{1 - 2r\cos(\theta - \varphi) + r^2}.$$

由 $\cos(\theta - \varphi) = \mp \sin \varphi$，代入 $1 - r^2$ 与分母化简后直接得到

$$P_r(\theta - \varphi) = \frac{\sin \pi y}{1 - \cos \pi y \sin \varphi}.$$

(b) 做变量替换 $t = F(e^{i\varphi})$，则

$$e^{i\varphi} = \frac{i - e^{\pi t}}{i + e^{\pi t}}.$$

取虚部并化简得

$$\sin \varphi = \frac{1}{\cosh \pi t}.$$

两边对 t 求导得

$$\frac{d\varphi}{dt} = \frac{\pi}{\cosh \pi t}.$$

代入泊松积分

$$\frac{1}{2\pi}\int_0^{\pi} P_r(\theta-\varphi)\tilde{f}_0(\varphi)\,d\varphi = \frac{1}{2\pi}\int_0^{\pi}\frac{\sin \pi y}{1-\cos \pi y \sin \varphi}\tilde{f}_0(\varphi)\,d\varphi.$$

替换变量 $\varphi \mapsto t$，积分区间变为 $(-\infty, +\infty)$，代入 $\sin \varphi$ 与 $d\varphi$ 即得

$$= \frac{\sin \pi y}{2}\int_{-\infty}^{\infty}\frac{f_0(t)}{\cosh \pi t - \cos \pi y}\,dt.$$

(c) 对 $\varphi \in (-\pi, 0)$，同理做变量替换 $t = F(e^{i\varphi}) - i$，有 $e^{i\varphi} = \dfrac{i + e^{\pi t}}{i - e^{\pi t}}$，可得

$$\sin\varphi = -\frac{1}{\cosh \pi t}, \quad \frac{d\varphi}{dt} = -\frac{\pi}{\cosh \pi t}.$$

代入积分并翻转上下限消去负号，得到

$$\frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta - \varphi)\tilde{f}_1(\varphi)\, d\varphi = \frac{\sin \pi y}{2}\int_{-\infty}^{\infty} \frac{f_1(t)}{\cosh \pi t + \cos \pi y}\, dt.$$

10. Let $F :\to \mathbb{C}$ be a holomorphic function that satisfies

$$|F(z)| \le 1 \quad \text{and} \quad F(i) = 0.$$

Prove that

$$|F(z)| \le \left|\frac{z - i}{z + i}\right| \quad \text{for all } z \in \mathbb{H}.$$

证明 取单位圆盘到上半平面的共形映射

$$G(w) = i\frac{1 - w}{1 + w} : \mathbb{D} \to \mathbb{H},$$

其逆为 $G^{-1}(z) = \dfrac{i - z}{i + z}$，且 $G(0) = i$。

考虑复合函数 $F \circ G : \mathbb{D} \to \mathbb{C}$，由条件得

$$F \circ G(0) = F(i) = 0, \quad |F \circ G(w)| = |F(G(w))| \le 1,\ \forall w \in \mathbb{D}.$$

由 *Schwarz* 引理，有

$$|F \circ G(w)| = \left|F\left(i\frac{1 - w}{1 + w}\right)\right| \le |w|,\ \forall w \in \mathbb{D}.$$

令 $z = G(w)$，则 $w = G^{-1}(z) = \dfrac{i - z}{i + z}$，代入得

$$|F(z)| \le \left|\frac{i - z}{i + z}\right| = \left|\frac{z - i}{z + i}\right|,\ \forall z \in \mathbb{H}.$$

证毕。

11. Show that if $f : D(0, R) \to \mathbb{C}$ is holomorphic, with $|f(z)| \le M$ for some $M > 0$, then

$$\left|\frac{f(z) - f(0)}{M^2 - \overline{f(0)}f(z)}\right| \le \frac{|z|}{MR}.$$

[Hint: Use the Schwarz lemma.]

证明 构造

$$g(z) = \frac{\dfrac{f(Rz)}{M} - \dfrac{f(0)}{M}}{1 - \dfrac{\overline{f(0)}}{M}\dfrac{f(Rz)}{M}},$$

故而 $g : \mathbb{D} \to \mathbb{D}$ 且是一个圆盘自同构，并且有 $g(0) = 0$。由 *Schwarz* 引理即得 $|g(z)| \le |z|$，换元 $z \to Rz$，得

$$\left|\frac{f(z) - f(0)}{M^2 - \overline{f(0)}f(z)}\right| \le \frac{|z|}{MR}, \quad \forall z \in B(0, R).$$

得证

12. A complex number $w \in \mathbb{D}$ is a fixed point for the map $f : \mathbb{D} \to \mathbb{D}$ if $f(w) = w$.
(a) Prove that if $f : \mathbb{D} \to \mathbb{D}$ is analytic and has two distinct fixed points, then f is the identity, that is, $f(z) = z$ for all $z \in \mathbb{D}$.
(b) Must every holomorphic function $f : \mathbb{D} \to \mathbb{D}$ have a fixed point? [Hint: Consider the upper half-plane.]

证明 *(a)* 不妨设这两个不动点为 w_1, w_2

构造自同构 $\psi(z) = \frac{z-w_1}{1-\overline{w_1}z}$，令 $g = \psi \circ f \circ \psi^{-1}$，则 $g(0) = 0$。由 *Schwarz* 引理 $|g(z)| \le |z|$，且 g 有非零不动点 $\psi(w_2)$，故 $g(z) = z$，即 $f(z) = z$。

(b) 不一定。反例：取 $F(z) = z + 1 : \mathbb{H} \to \mathbb{H}$，通过共形映射 $G(w) = i\frac{1-w}{1+w}$ 构造 $f = G^{-1} \circ F \circ G : \mathbb{D} \to \mathbb{D}$，则 f 无不动点。

13. The pseudo-hyperbolic distance between two points $z, w \in \mathbb{D}$ is defined by

$$\rho(z, w) = \left| \frac{z - w}{1 - \overline{w}z} \right|.$$

(a) Prove that if $f : \mathbb{D} \to \mathbb{D}$ is holomorphic, then

$$\rho(f(z), f(w)) \le \rho(z, w) \quad \text{for all } z, w \in \mathbb{D}.$$

Moreover, prove that if f is an automorphism of $\mathbb{D}$ then f preserves the pseudo-hyperbolic distance

$$\rho(f(z), f(w)) = \rho(z, w) \quad \text{for all } z, w \in \mathbb{D}.$$

[Hint: Consider the automorphism $\psi_\alpha(z) = (z - \alpha)/(1 - \overline{\alpha}z)$ and apply the Schwarz lemma to $\psi_{f(w)} \circ f \circ \psi_w^{-1}$.]
(b) Prove that

$$\frac{|f'(z)|}{1 - |f(z)|^2} \le \frac{1}{1 - |z|^2} \quad \text{for all } z \in \mathbb{D}.$$

This result is called the Schwarz-Pick lemma. See Problem 3 for an important application of this lemma.

证明 *(a)* 令 $\psi_w(z) = \frac{z-w}{1-\overline{w}z}$，则 $\rho(z, w) = |\psi_w(z)|$。令 $h = \psi_{f(w)} \circ f \circ \psi_w^{-1}$，由 *Schwarz* 引理 $|h(\zeta)| \le |\zeta|$，即 $\rho(f(z), f(w)) \le \rho(z, w)$。

自同构时 $$ 反向不等式成立，故等号成立。

(b) 令 $w \to z$，对 $\rho(f(z), f(w)) \le \rho(z, w)$ 取极限，即得

$$\frac{|f'(z)|}{1 - |f(z)|^2} \le \frac{1}{1 - |z|^2}.$$

14. Prove that all conformal mappings from the upper half-plane″ to the unit disc $\mathbb{D}$ take the form

$$e^{i\theta} \frac{z - \beta}{z - \overline{\beta}}, \quad \theta \in \mathbb{R} \text{ and } \beta \in \mathbb{H}.$$

证明 标准共形映射为

$$G^{-1}(z) = \frac{i-z}{i+z} :\to \mathbb{D}.$$

单位圆盘 $\mathbb{D}$ 的所有自同构为

$$\phi_\alpha(w) = e^{i\theta}\frac{w-\alpha}{1-\overline{\alpha}w}, \quad \alpha \in \mathbb{D},\ \theta \in \mathbb{R}.$$

因此 $\to \mathbb{D}$ 的任意共形映射可表示为

$$f(z) = \phi_\alpha \circ G^{-1}(z).$$

代入得

$$f(z) = e^{i\theta}\frac{\dfrac{i-z}{i+z} - \alpha}{1-\overline{\alpha}\cdot\dfrac{i-z}{i+z}} = e^{i\theta}\frac{(i-z)-\alpha(i+z)}{(i+z)-\overline{\alpha}(i-z)}.$$

令

$$\beta = G(\alpha) = i\frac{1-\alpha}{1+\alpha} \in,$$

则可化简为

$$f(z) = e^{i\theta}\frac{z-\beta}{z-\overline{\beta}}.$$

15. Here are two properties enjoyed by automorphisms of the upper half-plane.
(a) Suppose Φ is an automorphism of′ that fixes three distinct points on the real axis. Then Φ is the identity.
(b) Suppose (x_1, x_2, x_3) and (y_1, y_2, y_3) are two pairs of three distinct points on the real axis with

$$x_1 < x_2 < x_3 \quad \text{and} \quad y_1 < y_2 < y_3.$$

Prove that there exists (a unique) automorphism Φ of″ so that $\Phi(x_j) = y_j$, $j = 1, 2, 3$. The same conclusion holds if $y_3 < y_1 < y_2$ or $y_2 < y_3 < y_1$.

证明 *(a)* 上半平面 $\mathbb{H}$ 的所有自同构都是实系数分式线性变换

$$\Phi(z) = \frac{az+b}{cz+d}, \quad a, b, c, d \in \mathbb{R},\ ad - bc > 0.$$

若 Φ 固定三个不同实点 x_1, x_2, x_3，则

$$\Phi(x_j) = x_j, \quad j = 1, 2, 3.$$

即分式线性变换 $\Phi(z) - z$ 有三个不同零点。

而非零分式线性变换至多有两个零点，故

$$\Phi(z) - z \equiv 0 \implies \Phi(z) = z.$$

(b) 对任意三点组 (x_1, x_2, x_3)，存在唯一分式线性变换将其映为 $(0, 1, \infty)$。设 Φ_1 将 $(x_1, x_2, x_3) \mapsto (0, 1, \infty)$，$\Phi_2$ 将 $(y_1, y_2, y_3) \mapsto (0, 1, \infty)$，则

$$\Phi = \Phi_2^{-1} \circ \Phi_1$$

是″的自同构，且满足

$$\Phi(x_j) = y_j, \quad j = 1, 2, 3.$$

若有两个这样的自同构，则它们的复合固定三点，由 *(a)* 必为恒等映射，故唯一。

16. Let

$$f(z) = \frac{i - z}{i + z} \quad \text{and} \quad f^{-1}(w) = i\frac{1 - w}{1 + w}.$$

(a) Given $\theta \in \mathbb{R}$, find real numbers a, b, c, d such that $ad - bc = 1$, and so that for any $z \in$

$$\frac{az + b}{cz + d} = f^{-1}\left(e^{i\theta} f(z)\right).$$

(b) Given $\alpha \in \mathbb{D}$ find real numbers a, b, c, d so that $ad - bc = 1$, and so that for any $z \in$

$$\frac{az + b}{cz + d} = f^{-1}\left(\psi_\alpha(f(z))\right),$$

with ψ_α defined in Section 2.1.

(c) Prove that if g is an automorphism of the unit disc, then there exist real numbers a, b, c, d such that $ad - bc = 1$ and so that for any $z \in$

$$\frac{az + b}{cz + d} = f^{-1} \circ g \circ f(z).$$

[Hint: Use parts (a) and (b).]

证明 *(a)* 直接计算

$$\begin{aligned} f^{-1}(e^{i\theta} f(z)) &= i\frac{1 - e^{i\theta} \cdot \dfrac{i - z}{i + z}}{1 + e^{i\theta} \cdot \dfrac{i - z}{i + z}} \\ &= i\frac{(i + z) - e^{i\theta}(i - z)}{(i + z) + e^{i\theta}(i - z)}. \end{aligned}$$

令 $e^{i\theta} = \cos\theta + i\sin\theta$，代入化简得

$$f^{-1}(e^{i\theta} f(z)) = \frac{\cos\dfrac{\theta}{2}\ z + \sin\dfrac{\theta}{2}}{-\sin\dfrac{\theta}{2}\ z + \cos\dfrac{\theta}{2}}.$$

故

$$a = \cos\frac{\theta}{2},\ b = \sin\frac{\theta}{2},\ c = -\sin\frac{\theta}{2},\ d = \cos\frac{\theta}{2},$$

且

$$ad - bc = \cos^2\frac{\theta}{2} + \sin^2\frac{\theta}{2} = 1.$$

(b) 令

$$\psi_\alpha(w) = \frac{w - \alpha}{1 - \overline{\alpha} w}.$$

则

$$f^{-1}(\psi_\alpha(f(z))) = i\frac{1-\dfrac{\dfrac{i-z}{i+z}-\alpha}{1-\overline{\alpha}\cdot\dfrac{i-z}{i+z}}}{1+\dfrac{\dfrac{i-z}{i+z}-\alpha}{1-\overline{\alpha}\cdot\dfrac{i-z}{i+z}}} = i\frac{(1-\overline{\alpha})(i+z)-(i-z-\alpha(i+z))}{(1-\overline{\alpha})(i+z)+(i-z-\alpha(i+z))}.$$

设 $\alpha = x+iy$，化简即有

$$f^{-1}(\psi_\alpha(f(z))) = \frac{\dfrac{y}{1-|\alpha|^2}z+\dfrac{x-1}{1-|\alpha|^2}}{\dfrac{1+x}{1-|\alpha|^2}z-\dfrac{y}{1-|\alpha|^2}},$$

因此，

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{1-|\alpha|^2}\begin{pmatrix} y & x-1 \\ 1+x & -y \end{pmatrix}.$$

(c) 由于 $g:\mathbb{D}\to\mathbb{D}$ 是圆盘自同构，故而

$$g(z) = e^{i\theta}\psi_\alpha(z) = e^{i\theta}\frac{\alpha-z}{1-\overline{\alpha}z},$$

从而有 $f^{-1}\circ g\circ f = (f^{-1}\circ e^{i\theta}\circ f)\circ(f^{-1}\circ\psi_\alpha\circ f)$，于是有

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{1-|\alpha|^2}\begin{pmatrix} y & x-1 \\ 1+x & -y \end{pmatrix}\begin{pmatrix} \cos\frac{\theta}{2} & \sin\frac{\theta}{2} \\ -\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix},$$

即为 $f^{-1}\circ g\circ f$ 对应的 $abcd$ 的系数。